

STANLEY CHEGE THUITA

OPEN-SOURCE BITCOIN PROTOCOL ENGINEER & SYSTEMS DEVELOPER

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MISSION & TARGET FOCUS

OPEN SOURCE GRANTS • FELLOWSHIPS • PROTOCOL DEVELOPMENT

Systems developer and open-source Bitcoin engineer with a B.Sc. in Mathematics from the University of Nairobi. Specializes in zero-dependency Go architecture, Lightning Network Daemon (LND) gRPC integrations, and sovereign financial infrastructure. Graduate of the Btrust Builders / Africa Free Routing / HRF Lightning Bootcamp and co-founder of BitDevs Kisumu. Actively pursuing full-time upstream Bitcoin Core and Lightning protocol contributions through open-source developer grants (Brink, Btrust, OpenSats, Spiral) and research fellowships.

TECHNICAL CAPABILITIES

ATS-OPTIMIZED CATEGORIZATION

- Bitcoin & Lightning Protocol:** Lightning Network Daemon (LND), LND gRPC client architecture (`SendPaymentSync`, `FeeLimit_Fixed`), BOLT-11 HRP invoice parser, LNURL-pay RFC resolution, Regtest cluster automation, `bitcoin-cli`, `OP_RETURN` transaction anchoring, WebLN, Cashu / Fedimint architectures.
- Languages & Frameworks:** Go (Zero-Dependency Architecture, Goroutines, Channels, `sync.Mutex`/`RWMutex`, `net/http`, `embed.FS`), Ruby on Rails 7.1 (Hotwire Turbo, Solid Queue), TypeScript / React 18, Node.js, SQL (PostgreSQL), Bash/Shell scripting.
- Applied Cryptography & Infrastructure:** RFC 6238 TOTP, AES-256-GCM AEAD, SHA-256, Ed25519 digital signatures, Docker containerization, Linux systems administration, Git upstream PR workflow (rebase, squash, code review).

OPEN SOURCE PROTOCOL CONTRIBUTIONS & UPSTREAM PULL REQUESTS

VERIFIABLE IMPACT

minmoto/payment-channels — Open Source Protocol Contributor Merged PRs #23, #24, #30 | June 2026 – Present

- Implemented** portable payment channel schemas mapping East African (PR #23) and Southern African (PR #24) cash corridors using UN M49 geographical classifications and ISO 3166-1/4217 currency standards.
- Authored** Ethiopian Telebirr payment channel specification (PR #22) with E.164 phone number validation and explicit automation enum separation (`PaymentChannelAutomation.Manual` vs enterprise payout models).
- Engineered** schema extensions for Western Sahara cash channels (PR #30) and continental cash corridor registries utilizing centralized factory constructors.

Btrust Builders & Lightning Ecosystem — Protocol Contributor Merged PRs #139, #186, #133 | June 2026 – Present

- Completed** intensive Addis Ababa Lightning Bootcamp sponsored by Btrust Builders, Africa Free Routing, and Human Rights Foundation (HRF), training on Lightning Network internals, routing gossip, and channel management.
- Contributed** to African developer milestone registries and open-source portfolio repositories across PR #139, PR #186, and PR #133.

LIVE DEPLOYMENTS & BITCOIN PROOF OF WORK

ALTRADITS.VERCEL.APP/#DEPLOYMENTS

BitDevs Kisumu Chapter — Co-Founder & Technical Facilitator April 2026 – Present | bitdevs-kisumu.vercel.app

- Co-founded** Western Kenya's Bitcoin developer community; organized and facilitated inaugural Socratic Seminar #1 at LakeHub Kisumu under the Chatham House Rule.
- Curated** technical seminar agendas covering Bitcoin Core Pull Requests, BIP proposals (BIP-340/341 Taproot), Layer-2 gossip routing, and onboarding African engineers to protocol contributions.

YeboBank & Yebente Payment Layer — Founder & Systems Architect June 2026 – Present | yebobank.vercel.app | yebente.onrender.com

- Architected** pure-Go community banking engine (**YeboBank**) and settlement layer (**Yebente**) executing dual-rail fiat-to-Lightning payment bridging across East African mobile money corridors (M-Pesa, Telebirr, Zaad).
- Implemented** custom LND gRPC client with `MacaroonCredential` PerRPCCredentials authentication, TLS x509 cert pool validation, and bounded fee limits (`FeeLimit_Fixed`) executing synchronized payments via `SendPaymentSync`.
- Engineered** custom BOLT-11 HRP invoice decoder parsing multiplier prefixes (`m`, `u`, `n`, `p`) into satoshis and HTTP client resolving LNURL-pay RFC endpoints with deterministic SHA-256 invoice hashing.

Buna Beta E-Commerce Gateway — Full-Stack & Trade Corridor Architect July 2026 – Present | buna-beta.vercel.app

- Built** high-throughput cross-border trade gateway with Ruby on Rails 7.1, Hotwire (Turbo Drive, Turbo Frames, Turbo Streams), Stimulus.js, and PostgreSQL, deployed via serverless containers (`Dockerfile.vercel`).
- Implemented** database-backed asynchronous background workers with Solid Queue processing Safaricom Daraja STK Push IPN webhooks, decoupling physical trade logistics from financial settlement layers.

AltSocial & TouchType — P2P Coordination & Zero-Dependency Go April 2026 – Present | altsocial-one.vercel.app

- Engineered** **AltSocial** in React 18, TypeScript, Vite, Node.js/Express, and Google Gemini API, integrating Bitcoin Lightning Network payment support for automated subscription billing.
- Built** **TouchType** and backend engines in pure Go standard library (zero external dependencies in `go.mod`), leveraging `sync.Mutex` concurrency, `embed.FS` asset packaging, and Docker containerization.

ProofChain & Authentication — Applied Cryptography & Bitcoin Core Regtest Automation August 2026 – Present | github.com/altradits

- Automated** local Bitcoin Core Regtest testbeds using shell automation and `bitcoin-cli`, scripting 101-block coinbase maturation and anchoring SHA-256 certificate hashes into blockchain transactions via `OP_RETURN` script outputs.
- Implemented** RFC 6238 Time-Based One-Time Password (TOTP) from first principles in Go using `crypto/hmac`, `crypto/sha1`, big-endian binary buffers, dynamic 4-bit truncation modulo 10^6 , and AES-256-GCM authenticated symmetric encryption.

ENGINEERING APPRENTICESHIP

EDUCATION & MATHEMATICS

Zone01, Kisumu April 2026 – Present **University of Nairobi** Graduated Dec 2021

Software Engineering Apprentice **B.Sc. in Mathematics (Second Class Upper)**

- Completed 158 structured Go challenges mastering concurrency, pointers, channels, and network protocols.
- Maintained strict Git hygiene, branch rebasing, and peer review rigor across distributed microservice and CLI projects.
- Configured Docker containerization across Linux server production environments.
- Rigorous Coursework:** Number Theory, Linear Algebra, Abstract Algebra, Discrete Mathematics, Probability & Statistics, and Graph Theory.
- Mathematical Foundations:** Strong theoretical grounding in modular arithmetic, finite fields, discrete structures, and algorithmic complexity.